

Lista 10 de Cálculo II - Jeann Victor Batista

1) a) $g_{xx}(0,2) = -1$, $g_{xy}(0,2) = 6$, $g_{yy}(0,2) = 1$

$$H = \begin{bmatrix} g_{xx}(a,b) & g_{xy}(a,b) \\ g_{yx}(a,b) & g_{yy}(a,b) \end{bmatrix} \quad g_{xy} = g_{yx}$$

$$D = g_{xx}(a,b) \cdot g_{yy}(a,b) - [g_{xy}(a,b)]^2$$

$$= g_{xx}(0,2) \cdot g_{yy}(0,2) - [g_{xy}(0,2)]^2$$

$$= (-1) \cdot 1 - 6^2$$

$$= -1 - 36 \rightarrow -1 - 36 = -37 \quad D < 0, (0,2) \text{ é um ponto}$$

b) $g_{xx}(0,2) = -1$, $g_{xy}(0,2) = 2$, $g_{yy}(0,2) = -8$

$$D = (-1) \cdot (-8) - 2^2$$

$$8 - 4 \rightarrow 4$$

$$D > 0 \text{ e } g_{xx}(0,2) < 0, (0,2) \text{ é mínimo local}$$

c) $g_{xx}(0,2) = 2$, $g_{xy}(0,2) = -3$, $g_{yy}(0,2) = 7$

$$D = 2 \cdot 7 - [-3]^2$$

$$= 14 - 9$$

$$D > 0 \text{ e } g_{xx}(0,2) > 0, (0,2) \text{ é mínimo local}$$

$$= 5$$

d) $g_{xx}(0,2) = 4$, $g_{xy}(0,2) = 6$, $g_{yy}(0,2) = 9$

$$D = 4 \cdot 9 - 6^2$$

$$36 - 36 \rightarrow 0$$

$$D = 0, \text{ teste inconclusivo}$$

2) a) $f(x,y) = x^2 + xy + y^2 + 3x - 3y + 4$

$$f_x = \frac{\partial}{\partial x} (x^2 + xy + y^2 + 3x - 3y + 4) = 2x + y + 3 \quad f_x = 0 \rightarrow 2x + y + 3 = 0$$

$$f_y = \frac{\partial}{\partial y} (x^2 + xy + y^2 + 3x - 3y + 4) = x + 2y - 3 \quad f_y = 0 \rightarrow x + 2y - 3 = 0$$

$$\begin{cases} 2x + y + 3 = 0 \\ x + 2y - 3 = 0 \end{cases} \quad 2 \cdot (-4) + y + 3 = 0 \rightarrow -8 + y + 3 = 0 \rightarrow y - 5 = 0 \rightarrow y = 5$$

$$x + 2 \cdot 5 - 3 = 0 \rightarrow x + 10 - 3 = 0 \rightarrow x + 7 = 0 \rightarrow x = -7$$

$$3x + 3y = 0 \rightarrow x + y = 0 \rightarrow x = -y \quad (-3, 3) \rightarrow \text{ponto crítico}$$

$$f_{xx} = \frac{\partial^2}{\partial x^2} (x^2 + xy + y^2 + 3x - 3y + 4) = 2 \quad f_{yy} = \frac{\partial^2}{\partial y^2} (x^2 + xy + y^2 + 3x - 3y + 4) = 2 \quad f_{xy} = \frac{\partial^2}{\partial x \partial y} (x^2 + xy + y^2 + 3x - 3y + 4) = 1$$

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$D = f_{xx}(a,b) \cdot f_{yy}(a,b) - [f_{xy}(a,b)]^2$$

$$= f_{xx}(-3,3) \cdot f_{yy}(-3,3) - [f_{xy}(-3,3)]^2$$

$$= 2 \cdot 2 - [1]^2$$

$$= 4 - 1$$

$D > 0$ e $f_{xx}(-3,3) > 0$ é um ~~minimo~~ mínimo local

$$= 3$$

$$f(-3,3) = (-3)^3 - 3 \cdot 3 \cdot 3 + 3(-3) - 3 \cdot 3 \cdot 4 = -5$$

-5 é o mínimo local em (-3,3)

$$b) f(x,y) = x^3y + 12x^2 - 8y$$

$$f_x = 2f/2x = 3x^2y + 24x \quad f_y = 2f/2y = x^3 - 8$$

$$\begin{cases} 3x^2y + 24x = 0 \\ x^3 - 8 = 0 \end{cases}$$

$$3 \cdot 2^2y + 24 \cdot 2 = 0 \quad 12y + 48 = 0$$

$$12y = -48$$

$$(2,-4)$$

$$y = -4$$

$$f_{xx} = 2f/2x (3x^2y + 24x) = 6xy + 24 \quad f_{yy} = 2f/2y (x^3 - 8) = 0 \quad f_{xy} = 2f/2xy (3x^2y + 24x) = 3x^2$$

$$f_{xx}(2,-4) = 6 \cdot 2 \cdot (-4) + 24 = -24 \quad D = (-24) \cdot 0 - 12^2 = -144$$

$D < 0$, (2,-4) é o ponto de sela.

$$f(x,y)(2,-4) = 2^3 \cdot (-4) = -32$$

$$f(2,-4) = 2^3 \cdot (-4) + 12 \cdot 2^2 - 8 \cdot (-4) = 248$$

$$c) f(x,y) = y^3 + 3x^2y - 6x^2 - 6y^2 + 2$$

$$\frac{\partial f}{\partial x} = 6xy - 12x \quad \frac{\partial f}{\partial y} = 3y^2 + 3x^2 - 12y$$

$$\frac{\partial f}{\partial x}$$

$$\frac{\partial f}{\partial y}$$

$$\begin{cases} 6xy - 12x = 0 \rightarrow 6x(y-2) = 0 \rightarrow x=0 \text{ ou } y=2 \\ 3y^2 + 3x^2 - 12y = 0 \rightarrow y^2 + x^2 - 4y = 0 \end{cases}$$

$$y^2 + x^2 - 4y = 0 \rightarrow y^2 - 4y + 4 = 0 \rightarrow (y-2)^2 = 0 \rightarrow y=2$$

$$y^2 + x^2 - 4y = 0 \rightarrow y^2 - 4y + 4 = 0 \rightarrow (y-2)^2 = 0 \rightarrow y=2$$

$$y^2 + x^2 - 4y = 0 \rightarrow y^2 - 4y + 4 = 0 \rightarrow (y-2)^2 = 0 \rightarrow y=2$$

pontos críticos: (0,0), (0,4), (2,2), (2,2).

$$f_{xx} = 2f/2x (6xy - 12x) = 6y - 12 \quad f_{yy} = 2f/2y (3y^2 + 3x^2 - 12y) = 3y - 12$$

$$f_{xy} = 2f/2xy (6xy - 12x) = 6x$$

$$H = \begin{bmatrix} 6y - 12 & 6x \\ 6x & 3y - 12 \end{bmatrix}$$

(0,0)

$$f_{xx}(0,0) = 6 \cdot 0 - 12 = -12 \quad D = (-12)^2 - 0^2 = 144$$

$$f_{yy}(0,0) = 6 \cdot 0 - 12 = -12 \quad D > 0 \text{ e } f_{xx} < 0, (0,0) \text{ é máximo local}$$

$$f_{xy}(0,0) = 6 \cdot 0 = 0$$

(0,4)

$$f_{xx}(0,4) = 6 \cdot 4 - 12 = 12 \quad D = 12^2 - 0^2 = 144$$

$$f_{yy}(0,4) = 6 \cdot 4 - 12 = 12 \quad D > 0 \text{ e } f_{xx} > 0, (0,4) \text{ é mínimo local}$$

$$f_{xy}(0,4) = 6 \cdot 0 = 0$$

~~(-2,2)~~

$$f_{xx}(-2,2) = 6 \cdot 2 - 12 = 0 \quad D = 0^2 - (-12)^2 = -144$$

$$f_{yy}(-2,2) = 6 \cdot 2 - 12 = 0 \quad D < 0, (-2,2) \text{ é ponto de sela}$$

$$f_{xy}(-2,2) = 6 \cdot (-2) = -12$$

(2,2)

$$f_{xx}(2,2) = 6 \cdot 2 - 12 = 0 \quad D = 0^2 - 12^2 = -144$$

$$f_{yy}(2,2) = 6 \cdot 2 - 12 = 0 \quad D < 0, (2,2) \text{ é ponto de sela}$$

$$f_{xy}(2,2) = 6 \cdot 2 = 12$$

$$d) f(x,y) = x^3 - 12xy + 8y^3$$

$$\frac{\partial f}{\partial x} = 3x^2 - 12y \quad \frac{\partial f}{\partial y} = -12x + 24y^2$$

$$\begin{cases} 3x^2 - 12y = 0 \rightarrow x^2 - 4y = 0 \rightarrow x^2 = 4y \\ -12x + 24y^2 = 0 \rightarrow -x + 2y^2 = 0 \rightarrow x = 2y^2 \end{cases} \quad \begin{cases} (2y^2)^2 = 4y \rightarrow 4y^4 = 4y \rightarrow y^4 - y = 0 \\ y(y^3 - 1) = 0 \rightarrow y = 0 \text{ ou } y = 1 \end{cases}$$

$$y = 0 : x = 2 \cdot 0^2 = 0 \quad \text{ponto crítico: } (0,0)$$

$$y = 1 : x = 2 \cdot 1^2 = 2 \quad \text{ponto crítico: } (2,1)$$

$$f_{xx} = 2 \cdot 12x \quad (3x^2 - 12y) = 6x \quad f_{yy} = 2 \cdot 12y \quad (-12x + 24y^2) = 48y \quad f_{xy} = 2 \cdot 12 \quad (-12x + 24y^2) = -12$$

$$H = \begin{bmatrix} 6x & -12 \\ -12 & 48y \end{bmatrix} \quad D = 6x \cdot 48y - (-12)^2 = 288xy - 144$$

(0,0)

$$f_{xx}(0,0) = 6 \cdot 0 = 0 \quad D = 288 \cdot 0 \cdot 0 - 144 = -144$$

$$f_{yy}(0,0) = 48 \cdot 0 = 0 \quad D < 0, (0,0) \text{ é ponto de sela}$$

$$f_{xy}(0,0) = -12 \quad f(0,0) = 0 = 0 + 0 + 0$$

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(2,1)

$$f_{xx}(0,0) = 6 \cdot 0 = 0$$

$$D = 288 \cdot 0 \cdot 0 - 144 = -144$$

$$f_{xy}(0,0) = 48 \cdot 0 = 0$$

$D = 0$, $(0,0)$ é ponto de sela

$$f_{xy}(0,0) = 0$$

$$f(0,0) = 0 \cdot 0 \cdot 0 = 0$$

(2,1)

$$f_{xx}(2,1) = 6 \cdot 2 = 12$$

$$D = 288 > 0 = 144 - 144$$

$$f_{xy}(2,1) = 48 \cdot 1 = 48$$

$D > 0$ e $f_{xx} > 0$, $(2,1)$ é mínimo local

$$f_{xy}(2,1) = 48$$

$$f(2,1) = 8 \cdot 2 \cdot 1 = 16$$

$$p) f(x,y) = x \cdot e^{-2x^2-2y^2}$$

$$\frac{\partial f}{\partial x} = e^{-2x^2-2y^2} \cdot (1-4x^2) = -4x e^{-2x^2-2y^2} + (1-4x^2) e^{-2x^2-2y^2}$$

$$\frac{\partial f}{\partial y} = -4xy e^{-2x^2-2y^2}$$

$$f_{xx} = \frac{\partial}{\partial x} (-4x e^{-2x^2-2y^2}) = -4 e^{-2x^2-2y^2} + (-4x) \cdot (-4x) e^{-2x^2-2y^2}$$

$$f_{xy} = \frac{\partial}{\partial y} (-4x e^{-2x^2-2y^2}) = -4x \cdot (-4y) e^{-2x^2-2y^2} = 16xy e^{-2x^2-2y^2}$$

$$H = \begin{bmatrix} e^{-2x^2-2y^2} (-4 + 16x^2) & 16xy e^{-2x^2-2y^2} \\ 16xy e^{-2x^2-2y^2} & e^{-2x^2-2y^2} (16x^2 - 4) \end{bmatrix}$$

(1/2, 0)

$$f_{xx}(1/2, 0) = e^{-1/2} \cdot (-4 + 16 \cdot 1/4) = 2e^{-1/2}$$

$$D = (2e^{-1/2}) \cdot (2e^{-1/2}) = 8e^{-1}$$

$$f_{xy}(1/2, 0) = 0$$

$D > 0$ e $f_{xx} > 0$, $(1/2, 0)$ é máximo local

$$f_{xy}(1/2, 0) = 0$$

$$f(1/2, 0) = 1/2 e^{-1/2}$$

(-1/2, 0)

$$f_{xx}(-1/2, 0) = e^{-1/2} \cdot (-4 + 16 \cdot 1/4) = 2e^{-1/2}$$

$$D = (2e^{-1/2}) \cdot (2e^{-1/2}) = 8e^{-1}$$

$$f_{xy}(-1/2, 0) = 0$$

$D > 0$ e $f_{xx} > 0$, $(-1/2, 0)$ é mínimo local

$$f_{xy}(-1/2, 0) = 0$$

$$f_{xy}(-1/2, 0) = 0$$

$$f(-1/2, 0) = -1/2 e^{-1/2}$$

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$$f) f(x,y) = (x^2 + y^2) \cdot e^{y^2 - x^2}$$

$$\frac{\partial f}{\partial x} = 2x \cdot e^{y^2 - x^2} + (x^2 + y^2) \cdot e^{y^2 - x^2} \cdot (-2x) \cdot e^{y^2 - x^2} = 2x \cdot e^{y^2 - x^2} [1 - (x^2 + y^2)]$$

$$\frac{\partial f}{\partial y} = 2y \cdot e^{y^2 - x^2} + (x^2 + y^2) \cdot e^{y^2 - x^2} \cdot 2y \cdot e^{y^2 - x^2} = 2y \cdot e^{y^2 - x^2} [1 + (x^2 + y^2)]$$

$$\begin{cases} e^{y^2 - x^2} [2x - 2x(x^2 + y^2)] = 0 \rightarrow 2x - 2x(x^2 + y^2) = 0 \rightarrow 2x(1 - x^2 - y^2) = 0 \\ e^{y^2 - x^2} [2y + 2y(x^2 + y^2)] = 0 \rightarrow y = 0 \end{cases}$$

$$x = 0 : y = 0 \quad (0,0)$$

$$x^2 + y^2 = 1 : y = 0 \rightarrow x = \pm 1 \quad (-1,0), (1,0)$$

$$f_{xx} = -4xy \cdot e^{y^2 - x^2} + 2x(1 - x^2 - y^2) \cdot 2y \cdot e^{y^2 - x^2} \cdot e^{y^2 - x^2} [-4xy + 2xy(1 - x^2 - y^2)]$$

$$f_{yy} = [2(1 + x^2 + y^2) + 2y(2y)] \cdot e^{y^2 - x^2} + 2y(1 + x^2 + y^2) \cdot 2y \cdot e^{y^2 - x^2} = e^{y^2 - x^2} [2 + 6y^2 + 2x^2 + 4y^2] (x^2 + y^2 + 1)$$

$$f(0,0)$$

$$f_{xx}(0,0) = e^0 \cdot 2 = 2 \quad D = 2 \cdot 2 - 0 - 0$$

$$f_{xy}(0,0) = 0 \quad D > 0, f_{xx} > 0, (0,0) \text{ é mínimo local}$$

$$f_{yy}(0,0) = e^0 \cdot 2 = 2 \quad f(0,0) = 0$$

$$(1,0)$$

$$f_{xx}(1,0) = e^{-1} [2 - 6 - 0 + 0] = -4e^{-1} \quad D = (-4e^{-1}) \cdot (4e^{-1}) = -16e^{-2}$$

$$f_{xy}(1,0) = 0 \quad D < 0, (1,0) \text{ é ponto de sela}$$

$$f_{yy}(1,0) = e^{-1} [2 + 0 + 2 + 0] = 4e^{-1} \quad f(1,0) = 1 \cdot e^0 = 1$$

$$(-1,0)$$

$$f_{xx}(-1,0) = e^{-1} [2 - 6 - 0 + 0] = -4e^{-1} \quad D = (-4e^{-1}) \cdot (4e^{-1}) = -16e^{-2}$$

$$f_{xy}(-1,0) = 0 \quad D < 0, (-1,0) \text{ é ponto de sela}$$

$$f_{yy}(-1,0) = e^{-1} [2 + 0 + 2 + 0] = 4e^{-1} \quad f(-1,0) = 1 \cdot e^0 = 1$$

3 a) $f(x,y) = 4 + x^3 + y^3 - 3xy$

$\frac{\partial f}{\partial x} = 3x^2 - 3y$ $\frac{\partial f}{\partial y} = 3y^2 - 3x$

$3x^2 - 3y = 0 \rightarrow x^2 - y = 0 \rightarrow x^2 = y$

$3y^2 - 3x = 0$

$3(x^2)^2 - 3x = 0 \rightarrow 3x^4 - 3x = 0 \rightarrow x^4 - x = 0 \rightarrow x(x^3 - 1) = 0$ $x = 0$ ou $x = 1$
 $y = 0$ ou $y = 1$

pontos críticos: $(0,0), (1,1)$

b) $f_{xx} = \frac{\partial}{\partial x} (3x^2 - 3y) = 6x$ $H = \begin{bmatrix} 6x & -3 \\ -3 & 6y \end{bmatrix}$

$f_{yy} = \frac{\partial}{\partial y} (3y^2 - 3x) = 6y$ $D = 6x \cdot 6y - (-3)^2$
 $f_{xy} = \frac{\partial}{\partial y} (3x^2 - 3y) = -3$ $= 36xy - 9$

$(0,0)$

$f_{xx}(0,0) = 6 \cdot 0 = 0$

$D = 36 \cdot 0 \cdot 0 - 9 = -9$

$f_{yy}(0,0) = 6 \cdot 0 = 0$

$D < 0$, $(0,0)$ é ponto de sela

$f_{xy}(0,0) = -3$

$(1,1)$

$f_{xx}(1,1) = 6 \cdot 1 = 6$

$D = 36 \cdot 1 \cdot 1 - 9 = 27$

$f_{yy}(1,1) = 6 \cdot 1 = 6$

$D > 0$ e $f_{xx} > 0$, $(1,1)$ é mínimo local

$f_{xy}(1,1) = -3$

c) Sim, é possível falar a localização e natureza dos pontos críticos.

O padrão de curvas de nível em $(0,0)$ indica um ponto de sela.

O padrão construído em $(1,1)$ indica um mínimo local.

2) a) $f(x,y) = x^2 - 2xy + 3y^2 - x$, $D = \{(x,y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1\}$

$$\frac{\partial f}{\partial x} = 2x - 2y - 1 \quad \frac{\partial f}{\partial y} = -2x + 6y$$

$$\begin{cases} 2x - 2y - 1 = 0 \\ -2x + 6y = 0 \end{cases} \rightarrow 2x = 3/4 \rightarrow x = 3/8$$

$$-2x + 6y = 0$$

$$2y = 1 \quad y = 1/2$$

$$f(3/8, 1/2) = (3/8)^2 - 2 \cdot 3/8 \cdot 1/2 + 3 \cdot (1/2)^2 - 3/8 = 9/64 - 3/8 + 3/16 - 3/8 = -3/8$$

$$12/64 - 3/8 - 3/8 \rightarrow (12 - 6 - 12)/64 = -6/64 = -3/8$$

Lado $x = 0$ ($0 \leq y \leq 1$): $f(0,y) = 0 - 0 + 3y^2 - 0 = 3y^2$

extremos: $y = 0: f(0,0) = 0$ // $y = 1: f(0,1) = 3$

Lado $x = 1$ ($0 \leq y \leq 1$): $f(1,y) = 1^2 - 2 \cdot 1 \cdot y + 3y^2 - 1 = 3y^2 - 2y$

$$d/dy (3y^2 - 2y) = 6y - 2 \rightarrow 6y - 2 = 0 \rightarrow 6y = 2 \rightarrow y = 1/3$$

Valores $y = 0: f(1,0) = 0$

$y = 1: f(1,1) = 3 - 2 = 1$

$y = 1/3: f(1, 1/3) = 3 \cdot (1/3)^2 - 2 \cdot 1/3 = 1/3 - 2/3 = -1/3$

Lado $y = 0$ ($0 \leq x \leq 1$): $f(x,0) = x^2 - x$

$$d/dx (x^2 - x) = 2x - 1 \rightarrow 2x - 1 = 0 \rightarrow 2x = 1 \rightarrow x = 1/2$$

Valores $x = 0: f(0,0) = 0$

$x = 1: f(1,0) = 0$

$x = 1/2: f(1/2, 0) = -1/4$

Lado $y = 1$ ($0 \leq x \leq 1$): $f(x,1) = x^2 - 3x + 3$

$$d/dx (x^2 - 3x + 3) = 2x - 3 \rightarrow 2x - 3 = 0 \rightarrow 2x = 3 \rightarrow x = 3/2$$

Valores: $x = 0: f(0,1) = 3$ // $x = 1: f(1,1) = 1$

$(0,0): f(0,0) = 0$ // $(0,1): f(0,1) = 3$ // $(1,0): f(1,0) = 0$ // $(1,1): f(1,1) = 1$

minimum = $-3/8$ maximum = 3

b) $f(x,y) = x^2 + y^2 + x^2y + y$, $D = \{(x,y) \mid |x| \leq 1, |y| \leq 1\}$

$$\frac{\partial f}{\partial x} = 2x + 2xy \quad \frac{\partial f}{\partial y} = 2y + x^2$$

$$\begin{cases} 2x + 2xy = 0 \rightarrow x + xy = 0 \rightarrow x(1+y) = 0 & x=0 \text{ ou } y=-1 \\ 2y + x^2 = 0 \end{cases}$$

$$x=0 : 2y=0 \rightarrow y=0$$

$$y=-1 : 2 \cdot (-1) + x^2 = 0 \rightarrow x^2 = 2 \quad x = \pm\sqrt{2}$$

pontos críticos : $(0,0)$: $f(0,0) = 0$

lado $x=1$ ($-1 \leq y \leq 1$) : $f(1,y) = 1^2 + y^2 + 1^2y + y = y^2 + y + 1$

$$d/dy (y^2 + y + 1) = 2y + 1 \rightarrow 2y + 1 = 0 \rightarrow 2y = -1 \rightarrow y = -1/2$$

Valores : $y = -1$: $f(1,-1) = 1^2 + (-1)^2 + 1^2 \cdot (-1) + (-1) = 5$

$y = 1$: $f(1,1) = 1^2 + 1^2 + 1^2 \cdot 1 + 1 = 4$

$y = -1/2$: $f(1,-1/2) = 1^2 + (-1/2)^2 + 1^2 \cdot (-1/2) + (-1/2) = 1/4 - 1/2 + 1 = 5/4$

lado $y=1$: ($-1 \leq x \leq 1$) : $f(x,1) = x^2 + 1^2 + x^2 \cdot 1 + 1 = 2x^2 + 2$

$$d/dx (2x^2 + 2) = 4x \rightarrow x=0$$

Valores : $x=0$: $f(0,1) = 0^2 + 1^2 + 0^2 \cdot 1 + 1 = 2$

$x=1$: $f(1,1) = 1^2 + 1^2 + 1^2 \cdot 1 + 1 = 4$

$x=-1$: $f(-1,1) = (-1)^2 + 1^2 + (-1)^2 \cdot 1 + 1 = 4$

lado $y=-1$: ($-1 \leq x \leq 1$) : $f(x,-1) = x^2 + (-1)^2 + x^2 \cdot (-1) + (-1) = x^2 - 1$

$$f(x,-1) = x^2 - 1$$

mínimo : -1

// máximo : 0

c) $f(x,y) = x^2 + y^2 - 2x$, D é a região fechada com vértices $(2,0)$

$$\frac{\partial f}{\partial x} = 2x - 2 \quad \frac{\partial f}{\partial y} = 2y$$

equação da reta entre $(2,0)$ e $(0,2)$: $(y-0)/(2-0) = (x-2)/(0-2) \rightarrow (y-0)/(2-0) = (x-2)/(-2) \rightarrow (y-0)/(2-0) = (2-x)/2$

$$(y-0)/(2-0) = (2-x)/2 \rightarrow (y-0)/(2-0) = (2-x)/2 \rightarrow y = 2-x$$

$$-2y = 2x - 2 \rightarrow -y = x - 1 \rightarrow y = -x + 1$$

equação da reta entre $(2,0)$ e $(0,2)$: $(y-0)/(2-0) = (x-2)/(0-2) \rightarrow (y-0)/(2-0) = (2-x)/2$

$$(y-0)/(2-0) = (2-x)/2 \rightarrow y = 2-x$$

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Segmento entre $(2,0)$ e $(0,2)$: $y = x-2$ ($0 \leq x \leq 2$)

$$f(x, x-2) = x^2 + (x-2)^2 - 2x = 2x^2 - 6x + 4$$

Valores: $x=0: f(0,2) = 4$

$x=2: f(2,0) = 0$

$x=3/2: f(3/2, -1/2) = 1/2$

c) Segmento entre $(0,2)$ e $(0,-2)$: $x=0$ ($-2 \leq y \leq 2$)

$$f(0, y) = y^2$$

Valores: $y=-2: f(0,-2) = 4 \rightarrow y=2: f(0,2) = 4$

$y=0: f(0,0) = 0$

mínimo = -1 e máximo = 4 .

⑤ a) $f(x,y) = x^2 + y^2$, $xy = 1$

$$G(x,y) = xy - 1 = 0 \quad // \nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = (2x, 2y) \quad // \nabla G = \left(\frac{\partial G}{\partial x}, \frac{\partial G}{\partial y} \right) = (y, x)$$

$$\begin{cases} f_x = 2 \cdot 9x \\ f_y = 2 \cdot 9y \\ G(x,y) = 0 \end{cases} \rightarrow \begin{cases} 2x = 2 \cdot y \\ 2y = 2 \cdot x \\ xy = 1 \rightarrow x = 1/y \end{cases} \quad \begin{aligned} 2 \cdot 1/y = 2 \cdot y \rightarrow 2/y = 2 \cdot y = 2 \cdot y^2 \cdot y \rightarrow 2 = 2y^3 \rightarrow y = \pm 1 \\ 2y = 2y^3 \cdot 1/y \rightarrow 2y = 2y^3 \rightarrow 2y^3 - 2y = 0 \rightarrow 2y(y^2 - 1) = 0 \rightarrow y = \pm 1 \end{aligned}$$

$y = \pm 1 \rightarrow x = \pm 1$

$f(1,1) = 1^2 + 1^2 = 2 \quad // \quad f(-1,-1) = (-1)^2 + (-1)^2 = 2$ mínimo global = 2

b) $f(x,y) = 3x + y$, $x^2 + y^2 = 10$

$$G(x,y) = x^2 + y^2 - 10 = 0$$

$$\nabla f = (3, 1)$$

$$\nabla G = (2x, 2y)$$

$$\begin{cases} 3 = 2 \cdot 2x \rightarrow 3 = 2 \cdot 2 \cdot \sqrt{10-y^2} \rightarrow \lambda = 3 / (2 \cdot \sqrt{10-y^2}) \\ 1 = 2 \cdot 2y \\ x^2 + y^2 = 10 \rightarrow x^2 = 10 - y^2 \rightarrow x = \sqrt{10-y^2} \end{cases}$$

$$1 = \frac{3}{2 \cdot \sqrt{10-y^2}} \cdot 2y \rightarrow \frac{3y}{\sqrt{10-y^2}} = 1 \rightarrow (3y)^2 = (\sqrt{10-y^2})^2 \rightarrow 9y^2 = 10 - y^2 \rightarrow 10y^2 = 10 + y^2 - 1 \rightarrow y = \pm 1$$

$y=1: x = \sqrt{10-1^2} = \sqrt{9} = 3$

$y=-1: x = -\sqrt{10-(-1)^2} = -\sqrt{9} = -3$

$f(3,1) = 3 \cdot 3 + 1 = 10$

máximo global: 10 .

$f(-3,-1) = 3 \cdot (-3) - 1 = -10$

mínimo global: -10 .



c) $f(x,y) = e^{xy}$, $x^3 + y^3 = 16$ // $G(x,y) = x^3 + y^3 - 16 = 0$

$\frac{\partial f}{\partial x} = e^{xy} \cdot y$, $\frac{\partial f}{\partial y} = e^{xy} \cdot x$ $\nabla f = (e^{xy} \cdot y, e^{xy} \cdot x)$ $\nabla G = (3x^2, 3y^2)$

$$\begin{cases} e^{xy} \cdot y = \lambda \cdot 3x^2 & \frac{e^{xy} \cdot y}{e^{xy} \cdot x} = \frac{\lambda \cdot 3x^2}{\lambda \cdot 3y^2} \rightarrow \frac{y}{x} = \frac{x^2}{y^2} \rightarrow y^3 = x^3 \\ e^{xy} \cdot x = \lambda \cdot 3y^2 \\ x^3 + y^3 = 16 \end{cases}$$

$x^3 + x^3 = 16 \rightarrow 2x^3 = 16 \rightarrow x^3 = 8 \rightarrow x = 2$ $y = 2$

$f(2,2) = e^{2 \cdot 2} = e^4$ máximo global: e^4

d) $f(x,y,z) = x^2 + y^2 + z^2$, $x + y + z = 12$ $g(x,y) = x + y + z - 12 = 0$

$\nabla f = (2x, 2y, 2z)$ $\nabla G = (1, 1, 1)$ $(\lambda | 2 \cdot 3 = 12 \rightarrow 3\lambda = 12 \rightarrow \lambda = 4$

$$\begin{cases} 2x = \lambda \rightarrow x = \lambda/2 \rightarrow x = 2 \\ 2y = \lambda \rightarrow y = 2 \\ 2z = \lambda \rightarrow z = 2 \end{cases}$$
 $f(2,2,2) = 2^2 + 2^2 + 2^2 = 12$

$$\begin{cases} x + y + z = 12 \end{cases}$$
 $\min \text{ global} = 12$

6) $f(x,y,z) = 400xy^2z^3$ // $G(x,y,z) = x^2 + y^2 + z^2 - 1 = 0$

$\nabla f = (400y^2z^3, 800xy^2z^3, 1200xy^2z^2)$

$\nabla G = (2x, 2y, 2z)$

$$\begin{cases} 400y^2z^3 = \lambda \cdot 2x & 400y^2z^3 / 800xy^2z^3 = \lambda \cdot 2x / \lambda \cdot 2y \rightarrow y/2x \rightarrow x/y = y^2/2x \rightarrow y = \pm x\sqrt{2} \\ 800xy^2z^3 = \lambda \cdot 2y \end{cases}$$

$$\begin{cases} 1200xy^2z^2 = \lambda \cdot 2z & 800xy^2z^3 = \lambda \cdot 2y \rightarrow \frac{2z}{3y} = \frac{y}{3} \rightarrow 2z^2 = 3y^2 \rightarrow z = \pm y\sqrt{3/2} \end{cases}$$

$$\begin{cases} x^2 + y^2 + z^2 - 1 = 0 & 1200xy^2z^2 = \lambda \cdot 2z \end{cases}$$

$$z = \pm (\pm x\sqrt{2})\sqrt{3/2} = \pm x\sqrt{3}$$

$$x^2 + 2x^2 + 3x^2 = 1 \rightarrow 6x^2 = 1 \rightarrow x = \pm 1/\sqrt{6}$$

$$y = \pm \sqrt{2}/\sqrt{6} = \pm 1/\sqrt{3}$$
, $z = \pm \sqrt{3}/\sqrt{6} = \pm 1/\sqrt{2}$

puntos críticos $(\pm \frac{1}{\sqrt{6}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{2}})$



$$t = 200 \cdot (\pm 1/\sqrt{6}) \cdot (1/3) \cdot (\pm 1/2\sqrt{2}) = 200/6\sqrt{3}$$

$$\text{max global } 200/6\sqrt{3} \quad \text{min global } -200/6\sqrt{3}$$

7) $f(x, y) = 4x^3 - 4xy + y^3$
 $x^2 + y^2 = 25 = G(x, y)$

$$\nabla f = (8x - 4y, -4x + 2y) \quad \nabla G = (2x, 2y)$$

$$\begin{cases} 8x - 4y = \lambda \cdot 2x \rightarrow 4x - 2y = \lambda x \\ -4x + 2y = \lambda \cdot 2y \rightarrow -2x + y = \lambda y \\ x^2 + y^2 = 25 \end{cases}$$

$$\lambda = \frac{4x - 2y}{x} = \frac{4 - 2y}{x} \quad \frac{4 - 2y}{x} = \frac{-2x + y}{y} \rightarrow 3 - \frac{2y}{x} + \frac{2x}{y} = 0$$

$$\lambda = \frac{-2x + y}{y} = \frac{-2x}{y} + 1 \quad 3xy - 2x^2 + 2x^2 = 0 \rightarrow 2x^2 + 3xy - 2y^2 = 0$$

$$x = \frac{-3y \pm \sqrt{(3y)^2 - 4 \cdot 2 \cdot (-2y^2)}}{4} = \frac{-3y \pm \sqrt{9y^2 + 16y^2}}{4} = \frac{-3y \pm 5y}{4}$$

$$x' = 2y/4 = y/2 \quad x = y/2: (y/2)^2 + y^2 = 25 \rightarrow y^2/4 + y^2 = 25 \rightarrow 5y^2/4 = 25 \rightarrow y^2 = 20 \rightarrow y = \pm 2\sqrt{5}$$

$$x'' = -8y/4 = -2y \quad x = y/2: \pm \sqrt{5}$$

$$x = -2y: (-2y)^2 + y^2 = 25 \rightarrow 4y^2 + y^2 = 25 \rightarrow 5y^2 = 25 \rightarrow y = \pm \sqrt{5}$$

$$x = -2y: \pm 2\sqrt{5}$$

Pontos críticos: $(\sqrt{5}, 2\sqrt{5}), (-\sqrt{5}, -2\sqrt{5}), (-2\sqrt{5}, \sqrt{5}), (2\sqrt{5}, -\sqrt{5})$

$$(\sqrt{5}, 2\sqrt{5}): f = 4(\sqrt{5})^3 - 4\sqrt{5} \cdot (2\sqrt{5}) + (2\sqrt{5})^3 = 20 - 40 + 20 = 0$$

$$(-2\sqrt{5}, \sqrt{5}): f = 4(-2\sqrt{5})^3 - 4(-2\sqrt{5}) \cdot \sqrt{5} + (\sqrt{5})^3 = -80 + 40 + 5 = -35$$

$$(-\sqrt{5}, -2\sqrt{5}) = 0 \quad \parallel (2\sqrt{5}, -\sqrt{5}) = 125 \quad \text{máximo global } 125 \quad \parallel \text{min} = 0 \quad \parallel \text{maior temp} = 121^\circ\text{C}$$

$$8 \text{ a) } P(x, y, z) = xyz$$

$$G(x, y, z) = x + y + z - 100 = 0$$

$$\nabla P = (yz, xz, xy) \quad \nabla G = (1, 1, 1)$$

$$\left(P\left(\frac{100}{3}, \frac{100}{3}, \frac{100}{3}\right) = \left(\frac{100}{3}\right)^3 \right)$$

$$\begin{cases} yz = 2 \\ xz = 2 \\ xy = 2 \\ x + y + z = 100 \end{cases}$$

$$x = y = z$$

$$3x = 100 \rightarrow x = 100/3$$

$$\therefore 100/3, 100/3, 100/3$$

$$b) V = xyz = 1000 \text{ cm}^3$$

$$S = 2(x + y + z)$$

$$\nabla S = (2(y+z), 2(x+z), 2(x+y))$$

$$\nabla G = (yz, xz, xy)$$

$$G(x, y, z) = xyz - 1000 = 0$$

$$\begin{cases} 2(y+z) = 2 \cdot yz \\ 2(x+z) = 2 \cdot xz \\ 2(x+y) = 2 \cdot xy \\ xyz = 1000 \end{cases} \quad \begin{aligned} \frac{y+z}{x+z} &= \frac{y}{x} \rightarrow xy + xz = xy + yz \rightarrow xz = yz \rightarrow x = y \\ \frac{x+z}{x+y} &= \frac{z}{y} \rightarrow xy + xz + zy \rightarrow xy = zy \rightarrow x = z \end{aligned}$$

$$x^3 = 1000 \rightarrow x = 10 \text{ cm}$$

$$S(10, 10, 10) = 2 \cdot 300 = 600 \text{ cm}^2$$

$$\therefore 10 \text{ cm} \times 10 \text{ cm} \times 10 \text{ cm}$$